

SolarGrid Demo Utility — Network Power Flow Engineering Limits

SG-NET-005

SOLARGRID AI — Synthetic Engineering Demonstration Corpus

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Document Title	SolarGrid Demo Utility — Network Power Flow Engineering Limits
Version	1.0
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Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY
Purpose	Define synthetic network voltage, thermal, contingency, and power-flow validation limits.

Demo Note: This document contains synthetic engineering requirements created for the SOLARGRID AI demonstration. All numerical values, thresholds, equipment ratings, and procedures are illustrative and must be replaced or validated by the deploying organization before operational use.

Use restriction: This document is intended only for RAG testing, semantic retrieval testing, SafetyAgent demonstration, engineering evidence demonstration, PlannerAgent integration testing, and Tool 16 testing. It is not intended for real-world grid operation.

Document Control

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Scope: Define synthetic network voltage, thermal, contingency, and power-flow validation limits.

Revision History

Revision	Date	Description	Author
1.0	2026-01-15	Initial synthetic demonstration release	SolarGrid Demo Utility — Engineering Department

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1. Purpose and Scope

This engineering limits document defines the synthetic network model and power-flow validation limits used by the SolarGrid Demo Utility demonstration. It covers bus voltages, transmission lines, transformers, thermal loading, contingency assumptions, overload detection, and voltage violation detection.

All ratings and thresholds are synthetic demonstration values. They are not actual network ratings and do not represent an approved grid code.

2. Network Topology

Bus	Voltage Level	Typical Voltage Band	Connected Elements
N1	North 230 kV	0.98–1.02 pu	L1, L2
N2	Central 230 kV	0.97–1.03 pu	L2, L3, T1
N3	Solar 132 kV collector	0.96–1.04 pu	L3, L4
N4	Load 132 kV	0.95–1.05 pu	L4, T2
N5	BESS 132 kV	0.96–1.04 pu	L3, T2

Primary power-flow paths are represented by L1 through L4, with transformers T1 and T2. The BESS connects at N5 and the solar collector connects at N3.

3. Bus Voltage Limits

NET-001 — Normal voltage range is 0.95–1.05 pu.

NET-002 — Alert voltage range is 0.93–1.07 pu.

NET-003 — A calculated bus voltage below 0.93 pu or above 1.07 pu is an emergency-range voltage violation.

Voltage Condition	Range	Required Treatment
Normal	0.95–1.05 pu	Accept if all other constraints pass
Alert low	0.93–0.95 pu	Review margin and reactive-power support
Alert high	1.05–1.07 pu	Review margin and reactive-power support
Emergency low	<0.93 pu	Corrective action / safety review
Emergency high	>1.07 pu	Corrective action / safety review

4. Transmission Line Limits

Line	Connection	Continuous Rating	Planning Alert
L1	North–Central 230 kV	120 MVA	100% / 90%
L2	Central–Solar 230 kV	150 MVA	100% / 90%
L3	Solar–BESS 132 kV	80 MVA	100% / 90%
L4	Solar–Load 132 kV	100 MVA	100% / 90%

NET-004 — The maximum continuous line loading is 100% of the synthetic line rating.

NET-005 — Line loading above 90% shall generate a planning alert and require margin review.

5. Transformer Limits

Transformer	Connection	Continuous Rating	Planning Alert
T1	Central 230/132 kV	180 MVA	100% / 90%
T2	BESS/Load 132/33 kV	90 MVA	100% / 90%

NET-006 — The maximum continuous transformer loading is 100% of the synthetic transformer rating.

NET-007 — Transformer loading above 90% shall generate a planning alert.

6. Contingency Assumptions

NET-008 — The demonstration contingency screen shall evaluate the loss of one selected transmission element or generator when the scenario explicitly requests contingency validation.

NET-009 — A contingency result shall be labeled separately from the base-case power-flow result.

Contingency analysis in this synthetic corpus is a software demonstration concept. It is not a substitute for a utility's approved reliability methodology.

7. Power-Flow Validation

NET-010 — A power-flow validation shall confirm convergence, bus-voltage compliance, line loading compliance, transformer loading compliance, and applicable contingency status.

NET-011 — A non-converged power flow shall not be treated as PASS.

NET-012 — A power-flow result with any unmitigated voltage or thermal violation shall not be treated as PASS.

8. Overload and Violation Detection

Overload detection compares calculated apparent-power loading with the applicable continuous rating. Voltage violation detection compares calculated per-unit voltage with the Normal and Alert bands.

Calculated Condition	Status	Example Action
Line at 87%	Normal	No mandatory corrective action
Line at 94%	Alert	Review margin and future forecast
Line at 103%	Violation	Corrective action / replan
Bus at 0.96 pu	Normal	No voltage violation
Bus at 0.94 pu	Alert	Review voltage support
Bus at 0.92 pu	Violation	Corrective action / safety review

9. Evidence Requirements

NET-013 — A plan that relies on network feasibility shall have current power-flow evidence corresponding to the proposed operating state.

NET-014 — If network evidence is stale, missing, or unverifiable for a material topology or dispatch change, validation shall return UNKNOWN for the affected network constraint.